

True / False (20 marks)

Answer True or False and justify briefly.

1. True or False: The correct answer to 'A creature that is incapable of giving birth to offspring that are living as they exit is the' is 'beaver'.
2. True or False: The correct answer to 'Which of the following memories uses one transistor and one capacitor as basic memory unit' is 'DRAM'.
3. True or False: The correct answer to 'as an evergreen matures and grows, it's leaves tend to' is 'increase in number'.
4. True or False: The correct answer to 'Which of the following is an important reason to advocate the process of desecuritization?' is 'To prevent the erosion of established civil liberties through the removal of special rights and the cessation of extraordinary measures.'
5. True or False: The correct answer to 'Which of the following is NOT a characteristic of a city in the developing world?' is 'Produces a large share of its country's GDI'.
6. True or False: The correct answer to 'An animal cell that is permeable to water but not salts has an internal NaCl concentration of 10%. If placed in freshwater the cell will' is 'swell and eventually lyse'.
7. True or False: The correct answer to 'At first Descartes supposes that everything he sees is ____.' is 'FALSE'.
8. True or False: The correct answer to 'Which of the following processes occur in the cytoplasm of an eukaryotic cell? I. DNA replication II. Transcription III. Translation' is 'III only'.
9. True or False: The correct answer to 'The pull of gravity on Earth is caused by' is 'the mass of Earth.'
10. True or False: The correct answer to 'If the FED buys bonds in the secondary market' is 'the money supply will decrease.'

Long Form Response (20 marks)

Provide thorough reasoning and reference key steps.

1. Write a python function to count number of substrings with the sum of digits equal to their length.
2. Write a function to swap two numbers.

End of Paper